

Q 14

head = ① → ② → ③ → ④ → ⑤ → Null.

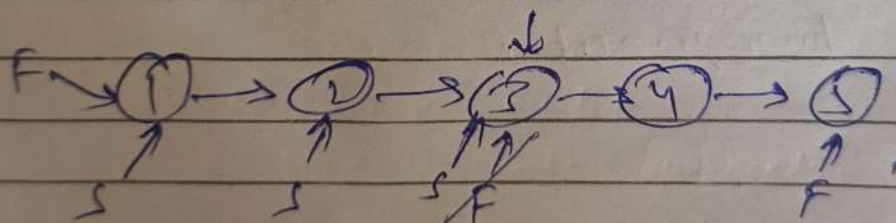
We have to reorder the linked list.

① → ④ → ② → ⑤ → ③ → Null

Concept:-

- ①. two part me split karo.
- ②. 2nd part ko reverse karo
- ③. Phir dono parts ko alternate position me merge kar do.

→ Slow-fast approach se do part me split karo.



ie

List 1 = ① → ② → ③

List 2 = ④ → ⑤

```

ListNode slow = head;
ListNode fast = head;
while (fast != null && fast.next != null) {
    slow = slow.next;
    fast = fast.next.next;
}

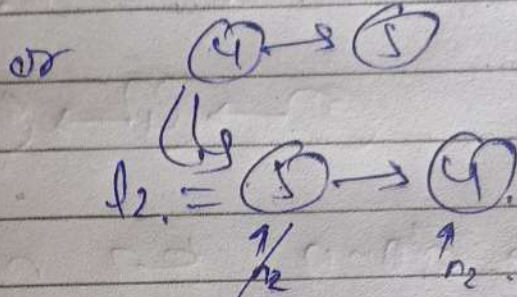
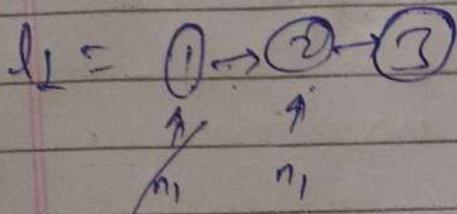
```

By this slow pointing the middle node.

// split

ListNode l2 = reverseList (slow.next);

slow.next = null;



Ab dono ko Alternate position me merge karenge.

while (l1 != null && l2 != null) {

 ListNode n1 = l1.next;

 ListNode n2 = l2.next;

 → Isme n1 me l1 ke next ko store karenge

 i.e. n1 = 2

 → Isme n2 me l2 ke next ko store karenge

 i.e. n2 = 4

 l1.next = n2; // l1 ke next me l2 ko point

 l2.next = n1; // l2 ke next me n1 ko point karenge.

 i.e. ① → ⑤ → ②

Phir

 l1 = n1; } dono ko aage

 l2 = n2; } badayenge.

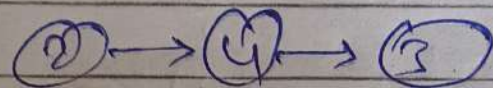
}

Ab

$n_1 = 3$

$n_2 = \text{null}$.

Phir same process karenge toh!



Ab dono ko merge karu do.

Private static ListNode reverseList(ListNode head) {

ListNode prev = null;

ListNode curr = head;

while (curr != null) {

ListNode next = curr.next;

curr.next = prev;

prev = curr;

curr = next;

}